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Urban performance at different boundaries in England and Wales through the settlement scaling theory

Hadi Arbabi[a], Martin Mayfield[b] and Gordon Dabinett[c]

ABSTRACT

The relationship between transport-led agglomeration and economic performance is evaluated in an English and Welsh context. We examine the effects of scale, i.e., inter- versus intra-city mobility infrastructure, on urban size–cost performance. An additional contribution of this paper lies in its use of power-law scaling models of urban systems, enabling an assessment of optimality in the trade-off between economic output and mobility costs accounting for ease of access within cities coupled with their built density. Findings suggest economic underperformance coincides with inadequate mobility at both inter- and intra-city scales, while overperformance is accompanied by overgrown urbanized area and escalating mobility costs.

KEYWORDS

agglomeration; urban scaling; scaling laws; urban performance; infrastructure planning

JEL O18, R00, R41
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INTRODUCTION

Cities and urban cores are the global nuclei of innovation, wealth generation, resource consumption and energy dissipation. Against a backdrop of expanding urbanized areas and increasing urban populations (Seto, Güneralp, & Hutyra, 2012) with limited resources available to sustain them, the need to design and maintain urban fabric and infrastructures in a manner that enables cities of higher productivity for minimal dissipated resources is evident. In practice, however, the ability to clearly identify, design and implement infrastructural measures, e.g., transport and mobility, that in fact improve economic performance, for a given definition of performance objective, is dependent on the availability of appropriate models and an understanding of the system at appropriate scales of intervention. This seemingly simple dependency could often suffer from the incongruities that may arise between the policy intentions, how they are interpreted into planning interventions, and the adequacy of the theoretical frameworks that are used to assess, inform and shape them.

These policy needs and the assortment of models and frameworks, abstract or otherwise, used to address them can be conceptualized along a disconnected spectrum of spatial scales. At the macro-scale end, broad national strategies could be thought of as shaped and informed by economic targets and demand pressure on the existing transport infrastructure. Such strictly rational formulation, however, would admittedly be at the mercy of political agendas and priorities. At the other intercity microscale end, the existing models and insights available are made up of the different variations on land-use and transport interaction (LUTI) models. These require a relatively long list of input parameters for calibration and operation, not many of which are routinely and homogeneously collected or easily available (Echenique, Grinevich, Hargreaves, & Zachariadis, 2013). Besides the inherent inability of these models to identify strategic infrastructural needs singlehandedly, their deployment at larger scales to inform mesoscale policies is hampered by the aforementioned data intensity and difficulties.

Infrastructural efforts at a mesoscale are thus driven by spatial agglomeration arguments whereby higher regional economic productivities are stimulated through the implementation of intercity mobility provisions in the image of those in Randstad, the Netherlands and

CONTACT

[a] **(Corresponding author)** harbabi1@sheffield.ac.uk
Department of Civil & Structural Engineering, University of Sheffield, Sheffield, UK.
[b] martin.mayfield@sheffield.ac.uk
Department of Civil & Structural Engineering, University of Sheffield, Sheffield, UK.
[c] g.e.dabinett@sheffield.ac.uk
Department of Urban Studies and Planning, University of Sheffield, Sheffield, UK.
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Germany's Rhine-Ruhr (Burger, Meijers, Hoogerbrugge, & Tresserra, 2015). The theoretical economic models currently used justifying these intercity mobility-enhancing strategies, however, remain in most parts inherently place non-specific with singular and arbitrary choices of spatial scales (Martin, Pike, Tyler, & Gardiner, 2015). While these frame the agglomeration process as a balance between increases in productivity and accumulating 'congestion costs', these size–cost balances remain abstract in such models (Abel, Dey, & Gabe, 2012).

The present paper therefore offers new insights into the effects of spatial scales on city performance balance and the extent to which this is influenced by the provision of transport infrastructure at different scales, i.e., intra- versus intercity mobility, using the urban system in England and Wales (EW) as an example. In order to do this, we adopt a scaling formulation of cities based on their population size with an explicit formulation of the balance between economic output, ease of mobility and its associated costs incurred with reference to the actual physical extent of cities and the larger urban systems to which they belong. To the best of the authors' knowledge, this paper is among the first instances of the application of such models when exploring explicit considerations of the city size–cost balance across spatial scales. Additionally, given that agglomeration frameworks and much of their evidence is based on Asian and North American urban systems, EW offer a particularly unique opportunity for the examination of these spatial effects within an urban system that is currently experiencing unique economic challenges (McCann, 2016, ch. 3) with an ever-widening divide that exists between the productivity and economic output of the south of England and the other regions in EW (Rowthorn, 2010).

Shaped and framed by the wider policy efforts stemming from the decentralization and devolution of certain powers to local entities in the form of combined authorities or city-regions (Gardiner, Martin, Sunley, & Tyler, 2013), the infrastructure policy debate in EW has been dominated by the attempts to address this performance gap. For national transport infrastructure, these attempts have generally been envisaged as creating and enabling these city-regions to act as single economic units by providing intercity transport infrastructures that reduce journey times, encouraging agglomeration economies (National Infrastructure Commission, 2016). The process of devising these transport links, inspired by intercity rail connectivity in the Randstad and Rhine-Ruhr, has only been argued at a singular spatial scale (Transport for the North, 2016, p. 63). Consequently, although these policies have been studied in terms of their implications for infrastructure governance and funding (O'Brien & Pike, 2015), an explicit exploration of the scale effects on the size–cost balances and performance in EW has largely been absent from both the policy papers and the larger academic debate.

The present results signal a systemic lack of adequate mobility and accessibility for a large portion of city units considered at various spatial scales, implying that lack of adequate mobility provisions is at the heart of a less-than-expected economic performance. While such effects are more easily noticeable at larger intercity scales, the problem is reoccurring at smaller scales and intra-city boundaries. This suggests that although intra-city transport-led agglomeration strategies are fitting, when implemented alone they would only mask mobility shortcomings at smaller scales without addressing the underlying causes of underperformance. As such, transport infrastructure planning cannot simply be led by agglomeration theory principles being applied at a single spatial scale and more concurrent consideration of urban scales is needed.

The paper is structured as follows. The next section provides the background to the scaling understanding of cities and introduces the specific model adopted in this study. The third section first provides a brief account of the methods and data used before establishing a broad empirical agreement with the theoretical scaling model and examining the average city mobility across spatial scales followed by their size–cost balance as compared with their idealized theoretical counterparts at these scales. In light of these comparisons, the fourth section discusses the implications of inadequate mobility provision and the potential of transport-led agglomeration at different spatial scales. Lastly, an extended account of boundary definitions and some additional results and larger figures are offered in Appendix A in the supplemental data online.

URBAN SCALING AND SETTLEMENT SCALING THEORY

In the past decade, with growing abilities to collect, share and analyse larger bodies of data pertaining to urban settlements, an understanding of cities and their properties as population scaling functions, formulated in the vein of similar allometric relations underlying the growth and size of organisms, has gained more traction both analytically and empirically (Bettencourt, Lobo, Helbing, Kühnert, & West, 2007). More importantly, this allometric line of thinking has already made an impression on the planning and economics literature. While part of this influence has been implicit in the form of concurrent observations of city rank–size distributions (Cheshire, 1999), others like Glaeser have been more explicit in the use of and reference to this field of literature, its theoretical frameworks and the models it provides in their own works (Glaeser, Ponzetto, & Zou, 2016). What is important from a policy and planning perspective is the ability of these frameworks to offer not only more tangible articulations of economic/energetic size–cost balances but also formulations that can be implemented and explored across a spectrum of scales without the obstacles faced by traditional LUTI models.

Bettencourt and West (2010) have put forward a notion of 'universal features' codifying the regularities present in various urban characteristics as a function of city population. They formalize these correlations of urban properties with city size as:

$$F(N) = F_0 N^{\beta} \tag{1}$$

where $F(N)$ denotes the average-aggregate urban characteristic of choice for population size N, e.g., gross value added (GVA), urbanized land area, employment etc.; F_0 is the baseline prevalence of F; and β is the exponent determining the nature of the scaling relation.

To explain these observations, a variety of urban models have been developed that yield such scaling behaviours for aggregated average response of urban attributes. These include models rooted in probabilistic conceptualizations of activities taking place in cities and the portion of the population contributing to them (Gomez-Lievano, Patterson-Lomba, & Hausmann, 2017) and those which are based on network realizations of the interactions between inhabitants and/or the geographical embedding of such networks within cities (Yakubo, Saijo, & Korošak, 2014).

Bettencourt (2013) introduces a simplified framework to derive such scaling behaviours that explicitly formulates a balance between the economic output generated in a city, Y, and the energy dissipated within the city, W, through the mobility processes that enable the population to generate this output. The simplest model, i.e., the social reactor model, from this Settlement Scaling theory starts from four simple underlying assumptions:

- The aggregate socioeconomic outputs are commensurate with the sum total of the number of human interactions locally.
- The population is mixing uniformly so that each individual has the access and minimum resources to explore the city (Jones, 2017).
- The infrastructure embedded in the city is a hierarchical network that grows incrementally and gradually keeping individuals connected with one another (Samaniego & Moses, 2008).
- The average baseline human production is not a function of population and remains constant across cities in an urban network (Szüle et al., 2014).

Before going any further, note that these assumptions, and in particular the second one, are to describe and model a highly idealized city. We will provide a discussion of the effects of their violation on the rest of the model behaviour at the end of this section. Nevertheless, based on the first two assumptions, the model formalizes an upper bound for average output for a city of population N as:

$$Y = \bar{g} a_0 l \frac{N^2}{A_n} \tag{2}$$

where $\bar{g}$ is the average strength of interactions between individuals; $a_0 l$ denotes the average area of coverage of individuals through which they experience the city; and N^2 is the total possible number of interactions over the entirety of the city's urbanized area, A_n. The product $\bar{g} a_0 l$, referred to as G hereafter, incorporates the extent of the mobility of individuals and their output. We note that it corresponds to the baseline human production mentioned in the fourth assumption and is presumed to be independent from population size, N. Although objections could be raised regarding a uniformly distributed series of individual interactions over a characteristic path validly portraying the mechanisms controlling encounters in and across all economic sectors and activities, the formulation is to be primarily that of a smoothed-out average of all behaviours rather than individual specific.

The second and third assumptions similarly enable a formalization of A_n in terms of population. Describing the geometry of the city and its inhabitants' average exploration path through it in terms of fractal dimensions, Bettencourt (2013) derives the scaling of the urbanized area as:

$$A_n = A_{n0} N^{1-\delta} \tag{3}$$

with:

$$\delta = H/(D(D+H))$$

where H and D are fractal dimensions characterizing the individual mobility paths and city geometry respectively; while A_{n0} embodies the baseline of the urbanized area and because of the second assumption would itself be a combined function of G and a notional average price of mobility per unit distance. Note that the real-life city geometries would imply a constraint of $2 \leq D \leq 3$ for city dimension, i.e., cities exist somewhere between a flat surface in space and the full volume of this surface extruded outwards. The individual's path, H, can more intuitively be thought of as the *ease of mobility* when travelling through the city. The full mixing assumption of Bettencourt's model would mean $H \approx 1$ in terms of the geometry of an inhabitant's access, effectively assuming that even an individual unfamiliar with the city fabric, e.g., its neighbourhoods and amenities, can afford to reach any place in the city. This encompasses both the availability of mobility modes connecting the city and their affordability to an average inhabitant.

The total mobility costs, W, over the urbanized extent can also be estimated by framing it as the energy dissipated through the hierarchy of the infrastructure modelled after parallel-connected resistors with electrical current through resistors replaced with the flow of people on roads etc. By substituting equation (3) back into equation (2), the scaling of economic output, its mobility costs and urbanized area can then be summarized as follows in a form similar to that of equation (1):[1]

$$\begin{cases} Y(N) = Y_0 N^{1+\delta} \\ W(N) = W_0 N^{1+\delta} \\ A_n(N) = A_{n0} N^{1-\delta} \end{cases} \tag{4}$$

where Y and W clearly scale super-linearly with population size through the exponent $\beta = 1 + \delta$, while urbanized area grows sub-linearly with $\beta = 1 - \delta$.

The social reactor model subsequently introduces the subtraction $Y - W$ as an indicator of a size–cost balance of cities. This objective function maximizes as a function of G at an optimal value, G^* (Figure 1). It is evident that the city is only viable when the balance between total output and mobility costs is positive, $G > G_{\min} = 0$, reaches a maximum size–cost balance at $G = G^*$, and becomes

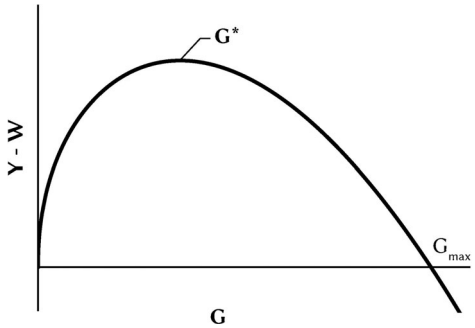

Figure 1. Schematic plot of economic output less mobility costs.

unstable again for $G_{\max} > G > G^*$ as the mobility costs start overwhelming beneficial gains. Also, for an idealized urban system where all cities strictly follow the scaling set out in equation (4), substituting the scaling of output and urbanized area back into equation (2) would provide an empirical estimate for the optimal baseline production as:

$$G^* \equiv \bar{g} a_0 l = A_{n0} \cdot Y_0 \tag{5}$$

where A_{n0} and Y_0 are the baseline prevalence of the urbanized area and economic output corresponding to the scaling of these attributes for an idealized city in a distribution.

The particular relevance of this model here, however, is the comparison opportunity it provides for gauging the state of this size–cost balance in cities and potential interventions that are implied to aid suboptimal performances. For cities where $G < G^*$, their true potential socioeconomic capacity is not reached. Looking back at the components of G, i.e., $\bar{g} a_0 l$, this is to say that the inhabitants are not fully exploring the city and hence cannot participate in as many interactions as the city has the potential to offer. As such, the interventions that would help increase the average individual areal coverage experiencing the city, e.g., better transport and accessibility within the city, would help the balance between the output and the energy used in the process of enabling people to generate it. For those with $G > G^*$, however, the socioeconomic success of the city is overshadowed by the escalating costs of the transport and mobility requirements embodied by the term W. Consequently, to tilt the scale back and lower mobility costs measures such as compaction of the city's built environment, its areal densification and higher efficiencies of the transport modes could be implemented.

Finally, it was acknowledged above that the assumption of a city with a uniformly mixing population with full and homogenous access to the city is a particularly strong one. For a simple idealized city, this assumption invariably results in presuming the city is two-dimensional ($D = 2$, i.e., the mobility processes required to bring inhabitants together take place on the ground over the city's flat surface rather than through its height and hence volume) and that individuals on average have full linear access to the city ($H = 1$). Carrying the assumption forward results in $\delta = 1/6$ and subsequent idealized expectations of the power-law exponent $\beta = (7/6), (7/6), (5/6)$ for the scaling of output, mobility costs and the urbanized area respectively in equation (4). Where cities become less fully accessible, H would take values less than unity, implying that inhabitants' experience of the city becomes restricted to disparate patches, e.g., neighbouring but virtually disconnected communities, within the city rather than linearly accessible extents. This would in turn shrink the value of δ and hence result in a scaling of Y and A_n closer to linear in equation (4) without violating outright any other aspect or framing of the overall model and/or the size–cost function.

SCALING AND SIZE–COST BALANCE

As discussed, this scaling formulation of cities can be used to offer categorical mesoscale comparisons of cities and urban regions against both an idealized realization of cities, i.e., $D = 2$ and $H = 1$, and any specific performance balance as observed in a particular city, e.g., London. This section starts by demonstrating the extent of the agreement between the underlying assumptions and resulting predictions from Bettencourt's model by estimating the scaling exponents for the GVA and urbanized land area for a number of different city boundary delineations. It then estimates values of G for each city following:

$$G = \frac{Y \cdot A_n}{N^2} \tag{6}$$

which is a rearrangement of equation (2). A comparison of these estimates for cities within each boundary definition with the optimal G^* calculated for their idealized fully accessible counterparts is then presented with an examination of the infrastructural needs of cities at different spatial scales.

Boundary scenarios

We conduct the analysis for 11 different realizations of city boundaries, seven of which are based on aggregating smaller cells with a minimum population density cut-off criterion. These are constructed based on the city clustering algorithm (CCA) described by Rozenfeld, Rybski, Gabaix, and Makse (2011). The algorithm continuously merges neighbouring cells, here on a square grid, based on the population density within cells until the neighbouring cells fall below a set density threshold. We have used the GEOSTAT grid (Office for National Statistics (ONS), 2016c) which contains population counts from the 2011 Census aggregated on a square grid of 1 × 1 km^2. A number of cut-off densities are then used within the range 100–3500 N/km^2 with the 1400 N/km^2 cut-off specifically adopted from Arcaute et al. (2015) identified as the threshold for well-defined cities providing a close match with major cities' urbanized extent.

For the administrative units, the local administrative unit level 1 (LAU-1) (ONS, 2016a) and the European Nomenclature of Territorial Units for Statistics level 3 (NUTS-3) (ONS, 2016b) are used. The LAU-1 and NUTS-3 units, however, become wholly inappropriate when considering London as they break down connected parts of the city to maintain statistical uniformity. As such for these boundaries, we have merged the smaller constituting units in accordance with the extent of the Greater London Authority (GLA). Finally, to include boundaries

theoretically consistent with the mixing population assumption in Bettencourt's (2013) idealized model which implies functional economies and labour markets, we use the travel-to-work area boundaries (TTWA) based on the commuting data from the 2011 Census (Coombes & ONS, 2015, p. 20).

Population, urbanized area and economic output estimation

The urban population for city units in each boundary definition was calculated by summing the population count from the GEOSTAT grid cells intersecting the city units of different boundaries. This means directly aggregating population counts from the grid in the case of the density-based boundaries and summing area-weighted counts for the other four boundaries based on the proportion of the area of the cells intersected by the city unit boundaries under a uniform cell-density assumption. Similarly, the urbanized area for each boundary was estimated by summing the area of the polygon segments from the contiguous built-up area layer intersecting boundary definitions.

The estimation of the economic output at different boundaries involves a few additional steps. We use the Organisation for Economic Co-operation and Development's (OECD) geographical information system (GIS)-based method using area-weighted proportionalities to arrive at the GVA estimates (OECD, 2012, pp. 45–48). First, the layer containing the GVA estimates, published by the ONS at NUTS-3 levels for 2011 (ONS, 2014), is intersected with the GEOSTAT population grid and each cell is assigned a portion of the GVA according to:

$$Y_{\text{cell}} = \sum \frac{Y_{\text{NUTS3}} \times ((N_{\text{cell}} \times A_i)/(A_{\text{cell}}))}{N_{\text{NUTS3}}} \qquad (7)$$

where Y_{cell} is the total GVA assigned to a cell in the population grid; N_{cell} and A_{cell} are the total population and area of the cell; Y_{NUTS3} and N_{NUTS3} are the GVA and population of the intersecting NUTS-3 areas respectively; and A_i is the area of the portion of the cell intersected by the corresponding NUTS-3 unit. Subsequently, a similar procedure is performed in reverse to aggregate back the GVA values from the population grid to the desired city-boundary definitions discussed previously.[2]

Urban performance in England and Wales

In obtaining the baseline prevalence and exponent of the scaling relations for urbanized area and GVA with population in each boundary definition, we use ordinary least squares (OLS) estimators on the linearized log transformation of equation (1). The larger numbers of the excessively small units, especially in urbanized area (UA) and C100 boundaries due to the smaller density cut-offs and small isolated built-up areas, however, would skew the tail of the power law and hence result in inappropriate linear fits. As such, arbitrary minimum population limits, e.g., 500,000 used by the OECD (2012) and 50,000 by Arcaute et al. (2015), are often applied to distinguish urban and metro areas from those that are rural.

Instead of using a single arbitrary limit across boundaries, we apply the statistical method described by Clauset, Shalizi, and Newman (2009), which estimates a lower bound in the distribution of empirical data above which a power-law distribution is followed to delineate 'urban' from 'rural' from a purely statistical perspective. This results in limiting the city units in each boundary to those following a rank–size population distribution (Rozenfeld et al., 2011) by finding the minimum population above which a power law could be assumed to apply (see Appendix A in the supplemental data online). These estimated minimum population cut-offs for each boundary definition and the OLS estimations for the units with populations above them are included in Table 1. While the overall regimes

Table 1. Summary of the boundary definitions used and the estimated scaling exponents β_{A_n} and β_Y.

Boundary		Units (N_{min})[a]	β_{A_n} [95% CI][f]	β_Y [95% CI]	R^2 β_{A_n}, β_Y
C100	Population density-based contiguous grid	586 (3895)	0.94 [0.92–0.95]	1.01 [1.00 1.02]	0.97, 0.98
C350		480 (7627)	0.94 [0.93–0.96]	1.02 [1.01 1.03]	0.98, 0.97
C500		103 (59,698)	0.96 [0.94–0.98]	1.02 [0.97 1.06]	0.99, 0.96
C750		111 (57,698)	0.96 [0.94–0.97]	1.02 [0.97 1.06]	0.99, 0.95
C1000		119 (55,031)	0.95 [0.93–0.97]	1.03 [0.98 1.07]	0.99, 0.95
C1400		96 (67,495)	0.96 [0.93–0.98]	1.03 [0.98 1.09]	0.99, 0.94
C3500		48 (66,671)	0.95 [0.92–0.99]	1.07 [1.00 1.14]	0.98, 0.95
UA[b]	Contiguous built-up area	1787 (1913)	0.94 [0.93–0.95]	1.00 [1.00 1.01]	0.91, 0.97
LAU-1[c,d]	Administrative	214 (101,355)	0.86 [0.82–0.91]	1.02 [0.96 1.08]	0.88, 0.84
NUTS-3[d]		34 (499,766)	0.79 [0.70–0.88]	1.29 [1.13 1.45]	0.90, 0.89
TTWA[e]	Functional economy	28 (510,149)	0.84 [0.76–0.91]	1.14 [0.99 1.29]	0.95, 0.91

Notes: [a]Values in parentheses denote the population cut-off for the smallest unit within each boundary definition when used for estimating the exponents.
[b]Urbanized area: based on the built-up area boundaries from December 2011.
[c]Local administrative units level 1.
[d]Constituting boundary units for the Greater London Authority (GLA) are aggregated and treated as one datum point in these boundaries instead of treating boroughs or Nomenclature of Territorial Units for Statistics level 3 (NUTS-3) boundaries as separate cities.
[e]Travel-to-work areas.
[f]CI, confidence interval.

for β_{A_n} and β_Y are in broad agreement with the expectations developed and observed by Bettencourt (2013) and Bettencourt and Lobo (2016), the OLS estimates for the majority of the boundaries for both properties fall much closer to unity. This is especially pronounced in the decreasing trend of β_Y estimates at larger scales as population density cut-off decreases.

These deviations from prescribed idealized values of the scaling exponents have previously been noted with exponents estimated for the UK lying much closer to unity rather than the expected values of 5/6 and 7/6 for sub- and super-linear scaling respectively (Arcaute et al., 2015). The larger matter of the comprehensiveness of these particular estimates is part of a broader ongoing debate that also includes issues around the appropriate methods of defining the boundaries of cities (Arcaute et al., 2016; Masucci, Arcaute, Hatna, Stanilov, & Batty, 2015). These, however, do not affect the present study since the derivation of the performance balance measure set out previously is independent of the estimated values of the exponents.[3] Within the framework of the social reactor model, however, this prevalent linear scaling can be interpreted as a sign that cities in EW on average exhibit a systemic pattern of impaired accessibility. As mentioned above, the extent of this lack of accessibility and mixing becomes increasingly larger at smaller population density cut-offs, e.g., 100 N/km^2, evident in the shrinking exponent estimates for the economic output. Nevertheless, the TTWA boundary estimates for GVA and urbanized land-area scaling exponents show a close match to those prescribed by the model, more or less appearing to uphold the mixing population assumption. Incidentally, these boundaries, which can be best described as self-contained functional economies (Coombes, 2010), also present the closest similarity in terms of definition to the idealized units modelled in the Settlement Scaling theory.

Figure 2 shows the estimates of G for individual city units across the boundary definitions against population on logarithmic axes. Despite the range of population size covered across the boundary definitions, the estimates of G remain more or less independent of population size ($(dG/dN) \approx 0$, $R^2 \in [0.00,\ 0.06]$) and within the same broad range across the different boundary definitions with an overall median of around 6.5E+6 £ m^2/N^2. The furthermost points to the right in each panel denote the different realizations of London and the GLA within each boundary definition. This regularity confirms the validity of the fourth assumption in Bettencourt's model and the model's broader relevance in the context of city units in EW.

Social reactor model theoretical optimum

To find the idealized optimal G^*, which maximizes the balance between economic output and the mobility costs of its generation, OLS fits with constant gradients in accordance

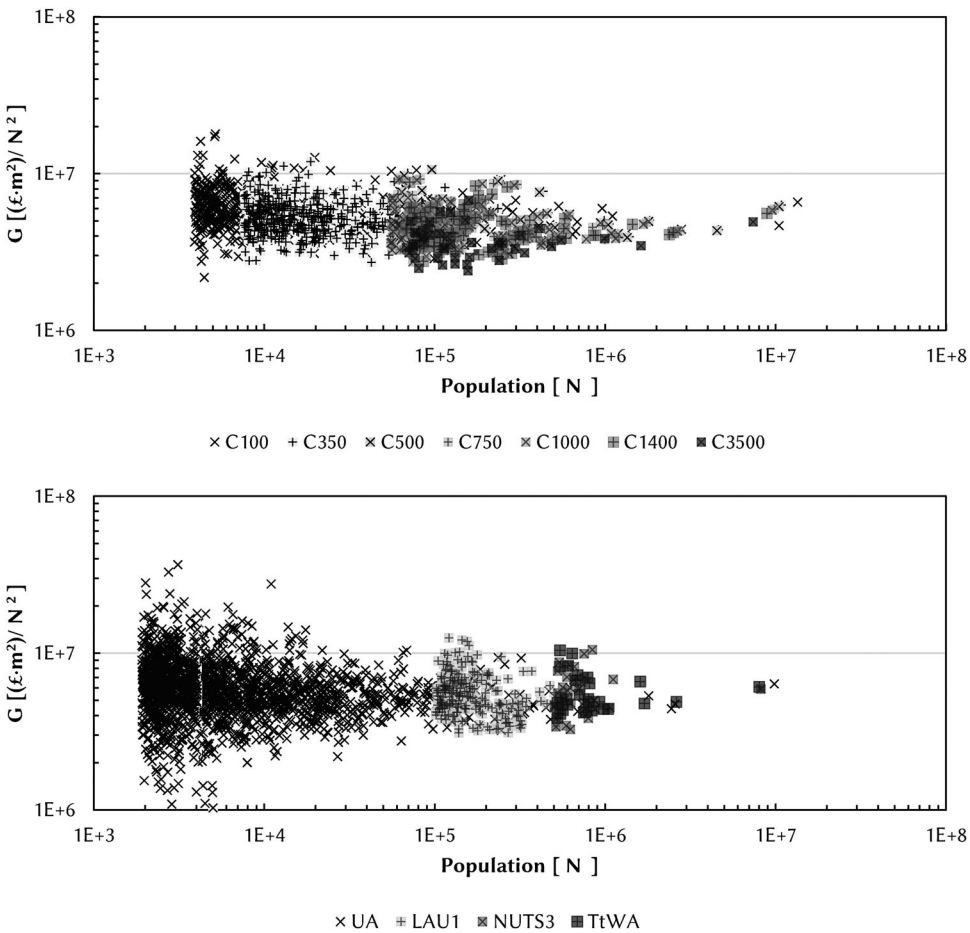

Figure 2. Superimposed plots of G against population for each boundary definition (note the logarithmic axes).

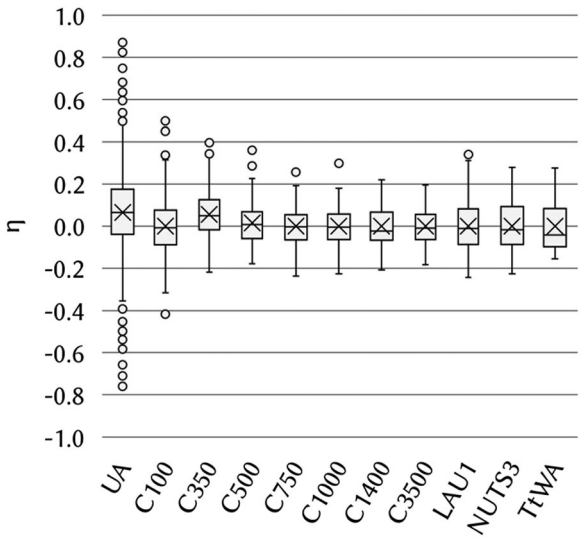

Figure 3. Box chart showing the distribution of $\eta(\equiv \log(G/G^*))$ within each boundary definition.

with those prescribed by the model, i.e., $\beta_{A_n} = (5/6)$ for urbanized area and $\beta_Y = (7/6)$ for the GVA, for idealized two-dimensional cities with full inhabitant mobility are implemented. We then take the product of the normalization coefficients from each fitted line, i.e., the intercepts A_{n0} and Y_0, to represent the theoretical optimal G^*, as set out in equation (5).

Figure 3 summarizes the distribution of the ratio (G/G^*), denoted as $\eta = \log(G/G^*)$, for city units in each boundary. At first glance, it is clear that estimates of G indeed tend to cluster close to the optimal value that maximizes the urban size–cost balance for idealized cities. A secondary observation can be made regarding the larger portion of the G estimates lying below the optimum highlighting a shortcoming in adequate levels of mobility and access in the city units across the boundary definitions used. This is more easily demonstrated by looking at the percentages of city units at different intervals of η where negative values indicate increasing lack of adequate mobility and mixing compared with the comparable idealized urban unit, while positive values indicate higher needs for increased built density (Figure 4). More than half the units in density-based boundaries with cut-offs > 750 N/km^2, the two administrative boundaries, and the TTWAs show ratios below the optimum.

A cursory inspection of the units in the UA and C350 boundaries, which exhibit larger portion of cities with $\eta > 0$, indicates this larger portion consists of city units often of a small population that are near larger units or in close proximity to a number of other similarly small units where the economic output is effectively not a product of the interactions within single individual units and would involve interactions and commutes between units or to larger nearby conurbations. This can be verified by estimating η for city units discarded previously with populations below the minimum cut-offs indicated in Table 1. For this comparison, we do not re-estimate the theoretical point of optimum anew but rather use the theoretical optimum obtained for the larger 'urban' units to quantify the notional performance balance of all city units compared with that of the average urban ideal (Figure 4, right panel).[4] This extension results in increases in the portion of units with larger-than-optimal η ratios, especially in boundaries that would include large numbers of small city units on the periphery of larger ones, i.e., UA, C100 and C350. The move from

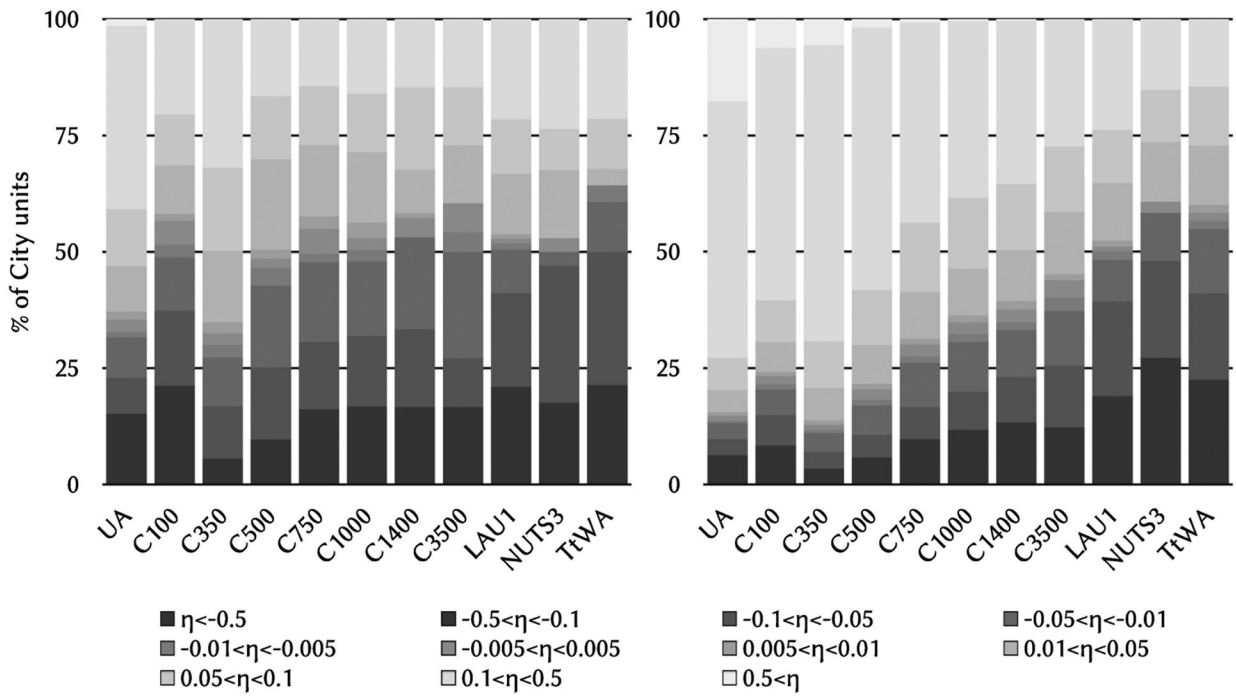

Figure 4. Bar charts showing the percentage of city units within the indicated range of η: (left) city units above the population cut-offs in Table 1; and (right) all city units.

the smaller density cut-offs in C100 to those in C3500 in essence eliminates the satellite commuter suburbs where, as mentioned, gains in GVA are not achieved over their own urbanized area.

Finally, we contextualize this optimality comparison geographically by mapping each boundary definition and the corresponding ratio estimates. Figures 5 and 6 illustrate these for C100, C500, C1000, C1400 and the other four non-density-based boundaries.[5] The maps show an estimated η for all city units within each boundary definition and not just those above the population cut-offs indicated in Table 1. The first visual pattern to be immediately evident, especially in the density-based boundaries, is the change from below-optimal ratios to those over the optimum crudely separating the south-east from the rest of EW. Another notable observation is that the units corresponding to Leeds and/or its greater city-region are the only major urban centres in the north exhibiting $G > G^*$ and, as such, the only northern urban core indicating a need for densification to improve

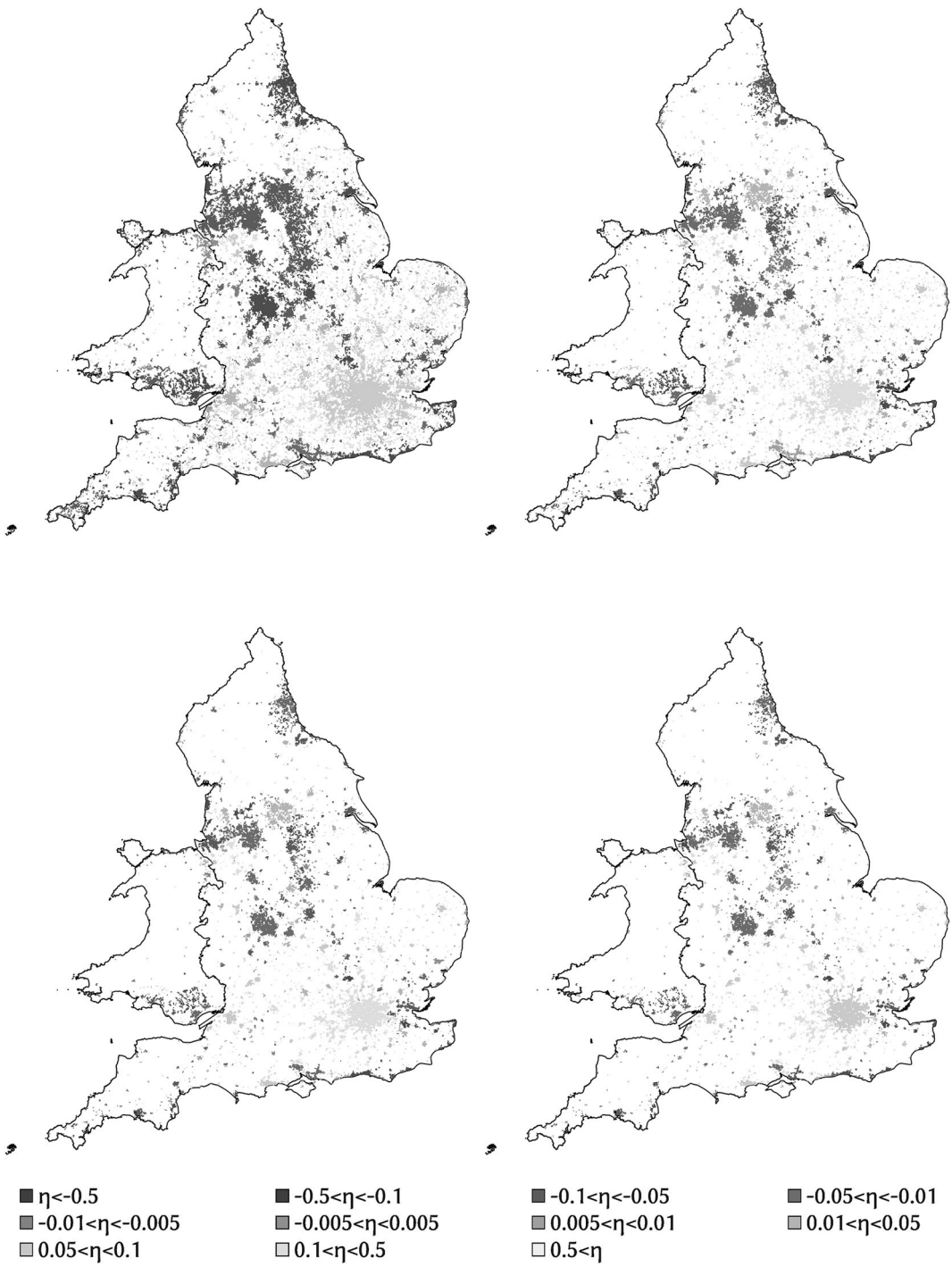

Figure 5. Maps of density-based boundaries colour coded based on the range of η: (top left) C100, (top right) C500, (bottom left) C1000 and (bottom right) C1400.
Sources: Office of National Statistics (ONS) and Ordnance Survey (OS) data © Crown copyright and database right 2017.

its size–cost balance rather than improvements to the intra-city transport similar to the rest. Additionally, subsequent disaggregation of the larger city unit of the north – note that in C100 the combined areas of Liverpool through to Leeds and then southwards through Nottingham are identified as a single city unit – in the density-based boundaries as the density cut-off is continually raised from 100 to 3500 N/km^2, does not appear to affect the identified size–cost balance where a need for better mobility persists, despite the changing scales. These remain largely stable even when comparing the corresponding boundaries in the non-density-based boundary definitions in Figure 6.

DISCUSSION

The planning policy in EW is being driven with an emphasis on connecting the underperforming cities through

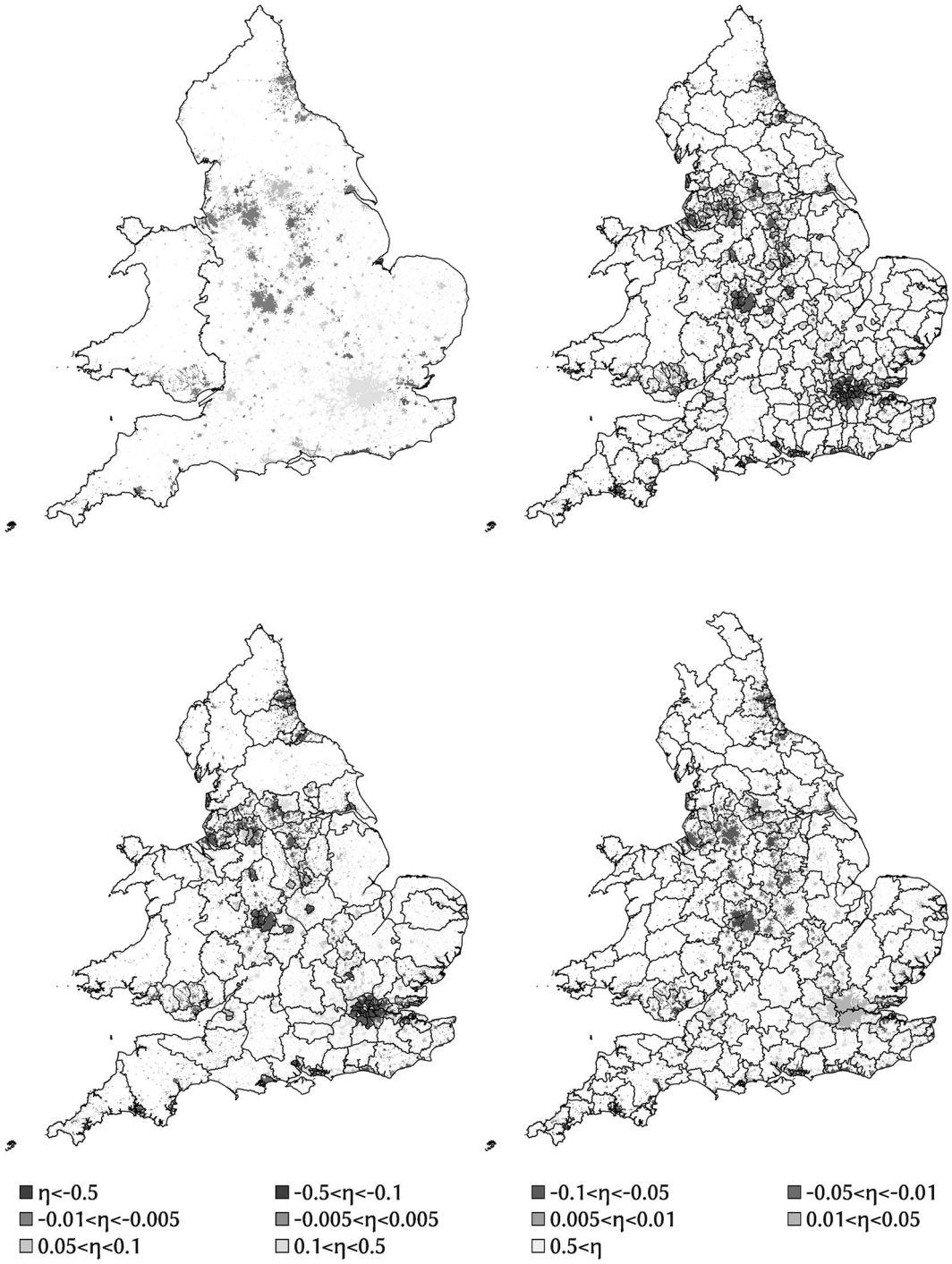

Figure 6. Maps of city units colour coded based on the range of η: (top left) urbanized area (UA), (top right) local administrative unit level 1 (LAU-1), (bottom left) Nomenclature of Territorial Units for Statistics level 3 (NUTS-3) and (bottom right) travel-to-work area (TTWA).
Sources: Office of National Statistics (ONS) and Ordnance Survey (OS) data © Crown copyright and database right 2017.

improved transport infrastructure. As mentioned, this is seen as fundamental at enabling these regions to perform as a single functional economy and as such contributing towards the rebalancing of the national economy (National Infrastructure Commission, 2016). These have precipitated in transport, specifically intercity connectivity, building up the largest portion of the infrastructure pipeline from 2017 onwards with project prioritization focused on reducing current travel times and reacting to the existing capacity demand while identifying the city-regions with the highest economic opportunity associated with their intercity connection (Infrastructure and Projects Authority, 2015).

The concluding observations from the previous section, however, noted a persistent lack of adequate mixing, or in other words a need for an improvement in the extent of the mobility provisions, in a majority of these regions regardless of the scale at which city-regions are considered from LAU-1s to larger TTWAs or density-based units. This is important when considering the generic recommendations borrowed from agglomeration theory regarding intercity transport policy. The overall transport and connectivity focus of such insights appears to be in agreement and supported by the social reactor model's interpretation of the current size–cost balance in EW across spatial scales. The intercity focus of stylized agglomeration principles, however, overlooks the overall performance balance, as formulated by Bettencourt (2013), and as such infrastructural needs across smaller-scale boundaries (Figures 5 and 6).

As an illustration, considering the C100 or C500 boundaries from Figure 5, centre-to-centre intercity transport links connecting Liverpool, Manchester, Sheffield and Leeds can be seen as beneficial. They would improve the performance balance as these regions appear as a single metro-region with an apparent lack of appropriate connectivity plaguing their size–cost performance that an intercity mobility scheme could potentially remedy. All the while, the individual incarnations of cities building up these areas in other boundaries, for the exception of Leeds, also show the same requirement for better mobility across smaller areal extents. This is indicative of a lack of accessibility at different levels starting from within the high-density core areas, e.g., those in C1400, and persisting at larger scales, e.g., those in C100 or TTWA.

With this in mind, transport-led agglomeration, as is often articulated as facilitating connectivity between major city centres, involves mobilizing populations into city units that may not individually have the transport capability to provide for the efficient mobility that is implicit in agglomeration theory and conducive to the improved size–cost performance of the overall aggregated regions. Although such single-scale interventions could perhaps increase economic output nominally, the size–cost analysis suggests that they would do so to the detriment of the overall comparative balance at other scales. In EW where the city centres have had the largest population growth in the last decade and accommodate the bulk of employment opportunities as the suburbs and rural areas provide the residential housing (Thomas, Serwicka, & Swinney, 2015, p. 30), multi-scale, i.e., intra- and intercity, infrastructural interventions provided concurrently would seem more coherent. Considering practicalities such as a limited funding bandwidth, prioritizing policy interventions to start from smaller scales and moving on to larger ones would adjust the size–cost performance more effectively since improved mobility at an intra-city scale facilitates intercity access while intercity access would only increase demand on existing intra-city infrastructure.

This is not to say that the intercity infrastructure is not needed or to imply that it constitutes an entirely wrong strategy. In fact, for any pair of cities where one exhibits $G > G^*$ and the other $G < G^*$, the model used in this study would project an estimate of G closer to G^* for the hypothetical and idealized city-region which would have the sum of the pair's population, urban extent and economic output. Thus, assuming the Leeds and Manchester areas comprise a well-connected metro-region, the model would project a better-balanced size–cost performance for this hypothetical city. There is, however, an implicit assumption in the model used here that the resulting aggregated metro-region is in itself a uniform urban conurbation providing for an ideal population mixing, meaning that for real-life examples a balancing of the performance of individual units would be required before a natural merging of the regions. In a sense, the agglomeration economies principle would perform as intended when the units are themselves performing well at smaller scales before a connection at larger scales so that the aggregation helps to introduce the efficiencies and productivities of higher populations. These effects will not inexplicably overcome the underperformance of the contributing cities if they are not previously addressed. This vital issue and importance of mobility at an intra-city scale is only often acknowledged in passing (National Infrastructure Commission, 2016).

On a slightly different note, one might question the degree to which approaching G^* is desirable and practical. Despite its more tangible formulation of what essentially are 'congestion costs', Bettencourt's model aggregates all costs associated with population mobility over the infrastructure network. The energy dissipated and the overall size–cost balance, G, then have to include, for instance, fuel/energy source, type and cost bundled together. In a context where most mobility solutions are fossil-fuel intensive and concerns for the effects of climate change exist, maximizing the economic output for the transport energy lost becomes an imperative. In such cases, G^* embodies this maximization point and target. However, for the same targets, policy could focus on decoupling modes of mobility and transport from their fuel sources instead. As an extreme illustration, if similar levels of mobility could be provided through freely available public transport run by renewable energy sources, the relevance of a $Y - W$ balance becomes diminished significantly. This would mean the optimum point of G^* may not be a practical target in situations where cities are indicating estimates more than the optimum since escalating mobility costs will not have the same tangibly negative implications.

In a broader context, this study also points to a different facet of the north–south divide in EW. This division of economic output can be reformulated with a size–cost perspective and seen as long-term planning needs. Despite higher economic outputs, economic success in a majority of the south-east appears to be achieved through mounting mobility costs as compared with idealized urban cities of the same population that would have exhibited smaller urbanized areas. Our examination of Bettencourt's explicit formulation of these balances shows that the cities in the south-east and London particularly have grown too large and require built-up area densification. This is in contrast to Cheshire's (2013) recent criticism of densification and urban containment strategies labelling them theoretically grounded but without empirical grounds. Moreover, Arcaute et al. (2016) use percolation at different distances on the UK road network to obtain a hierarchal classification of the road transport network and, by extension, the clustering of the geographical regions as represented by their road connectivity where the EW network initially collapses into one radiating out from London connecting the southern regions and the other connecting the north, Wales and Cornwall. A similar classification pattern can be observed in Figures 5 and 6 to some extent where the south-east exhibits an overwhelming need for more compactness, especially along the radiating motorways, while the majority of the north suffers from poor intra-city connectivity and mobility.

We would be remiss, however, if we did not point out the remaining shortcomings and potential direction. The majority of models from the same family of that used here start from the assumption that the units under study are in fact uniformly urban and functional economic catchments (Bettencourt, 2013; Gomez-Lievano et al., 2017; Yakubo et al., 2014). Unfortunately, this leaves them highly sensitive to the urban population count at each spatial scale and hence the choice of boundary used at that scale (Arcaute et al., 2015). While such fluctuations were observed in the G estimates from the model used in this study when considering slightly different boundaries at similar scales, the determination of the planning needs relative to the idealized city remained consistent. Louf and Barthelemy (2014) argue that until a comprehensive, universal and mechanistic understanding of how cities are formed evolve, and function is developed, such models should not be used in shaping policy advice. However, given that planning policy will be formed one way or the other and that the current economic agglomeration models informing policy not only suffer from the same fundamental lack of universality but also are (as previously mentioned) placeless and single scale in nature, providing and considering alternative pictures of city performance and infrastructural needs would benefit the overall policy and planning debate.

CONCLUSIONS

We have characterized the potential influence of spatial scales on overall city performance, especially when considering the provision of intra- or intercity transport solutions. A number of insights emerge from the study. Improvements of intercity mobility and access based on agglomeration principles are often argued for boosting productivity and economic performance. We have, however, demonstrated that the part played by inadequate mobility in regional economic underperformance appears to permeate through the change of spatial scales starting at smaller intra-city scales. In light of this, we strongly argue that planning transport infrastructure solutions at regional levels cannot only rely on the consideration of larger spatial scales alone and policy efforts should require an examination of planning needs across scales, particularly when larger-scale solutions could hide the root causes of unbalanced performance at scales below them.

Finally, we have demonstrated the use of Bettencourt's social reactor model, one amongst many within the data-driven science of cities, in identifying such regional infrastructural requirements that would maintain size–cost performance balance over multiple spatial scales in the context of the urban system in EW. Referring back to the disconnected conceptual spectrum of the infrastructural planning scales, we have shown the model's potential to provide a way of bringing together the macro-scale agendas with the microscale planning models at a mesoscale where the consistency of the planning strategies promoted by larger-scale concerns can be scrutinized across a variety of intermediate levels rather than a limited grouping of spatial data.

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DISCLOSURE STATEMENT

No potential conflict of interest was reported by the authors.

NOTES

1. Interested readers are encouraged to view the detailed derivation of the model based on a hierarchical urban infrastructure network and the optimization of output–energy balance in the supplemental data online of Bettencourt (2013).
2. Although not of importance to the focus of this paper, note that the simplicity of the OECD method might cause problems when aggregating back up to units not so larger than the base population layer resulting in linear scalings or noise recordings (Smith, 2014) due to the nature of the simple population proportionality and the uniform density distribution assumption in equation (7). For an

expanded discussion, see Appendix A in the supplemental data online.

3. The numerical value of the theoretically optimal G^* is not independent of the exponents observed for economic output and urbanized area. It is the overall maximization of $Y - W$ that does not depend on specific values of the exponents.

4. For OLS regression estimates of scaling exponents using all units in each boundary, see Appendix A in the supplemental data online.

5. For larger figures, see in Appendix A in the supplemental data online.

ORCID

Hadi Arbabi http://orcid.org/0000-0001-8518-9022
Martin Mayfield http://orcid.org/0000-0002-9174-1773

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